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Linear Regression

A statistical analysis tool used to investigate relationship between a dependent variable and one or more independent variable.

Simple Linear Regression

approximately Graphically it is a straight line. Linear Regression between exactly 1 independent and 1 dependent variable.

$$Y = \beta_0 + \beta_1 X + \epsilon$$

OUTPUT

$y = mx + c$
some what same as this

dependent variable that we want to predict

Y intercept. It is the default value of the model. Value of Y when $X = 0$.

It defines the slope of the line.

Independent variable.

random error margin

• error of $\bar{x}$ (mean value of x) will be 0

Some Assumptions that makes our life easy

• variance of error is constant across all values

• error of x_i doesn't effect error of x_{i+1}

• The change of error being +ve or -ve is same

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For Predicting and estimating our outcomes we made subtle changes to our expression

$$\hat{y} = b_0 + b_1 x$$

Annotations:

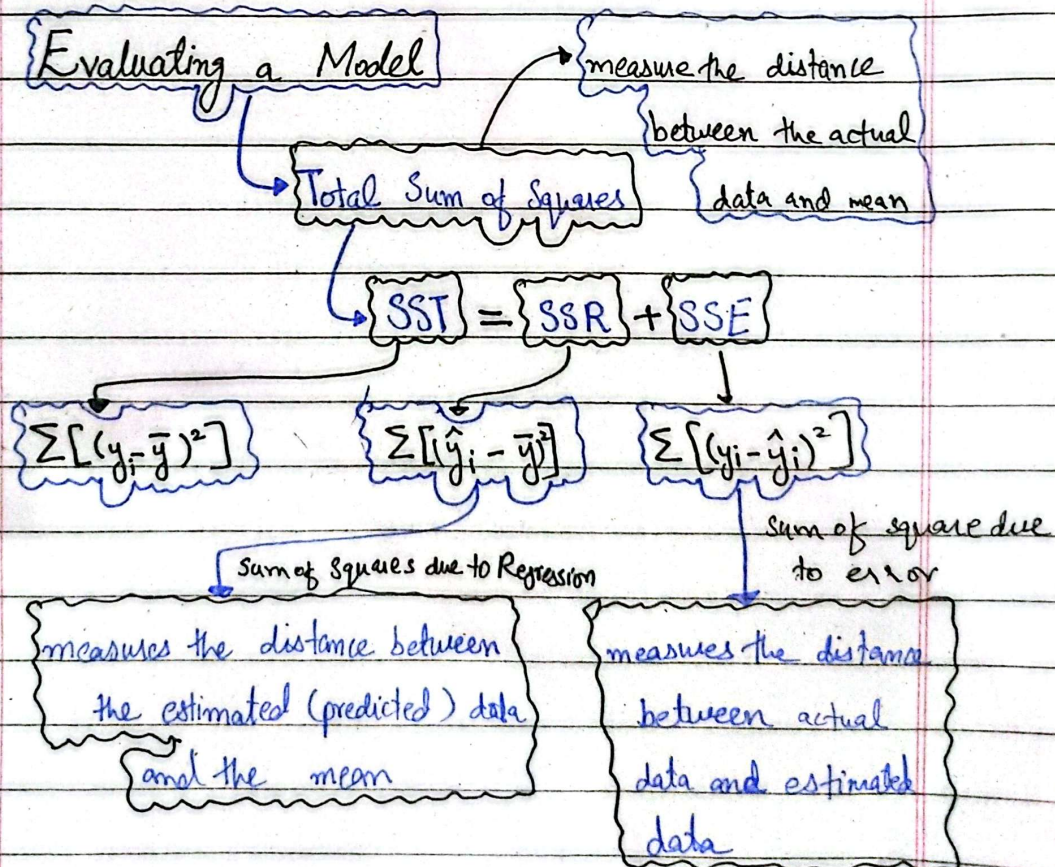
- $\hat{y}$: estimated output
- b_0 : estimated β_0
- b_1 : estimated β_1

In questions we always work with this formula

$$b_1 = \frac{\sum [(x_i - \bar{x})(y_i - \bar{y})]}{\sum [(x_i - \bar{x})^2]}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Evaluating a Model



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Coefficient of determination → Tells the strength of your model

$$r^2 = \frac{SSR}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

It tells how accurate your model is

ranges
between

0 to 1

1 means 100% accurate

0 means no accuracy at all

Sample Correlation Coefficient → tells the relation between x and y

$$r = + \text{ or } - \sqrt{r^2}$$

same as b_1 if b_1 is +ve then r is +ve and vice versa

-1 to 1

+ve → towards 1 means x and y are directly proportional

-ve → towards -1 means x and y are inversely proportional

values very close to 0 indicates no relation between x or y

